Worksheet 6 *Logarithms for fitting*

Prepares you for lab manual pages **Fit1, Fit2, Fit3, Fit4** (section L2 S6)

Name: Name of TA:

1. Suppose F satisfies the change equation $dF/dt = 0.3\,F$ and the initial condition $F(0) = 100$.

(a) Sketch a time series for F *on a semilog plot*, using natural logarithms (base e).



(b) If $F(t) = Ce^{kt}$, then $k =$ $C =$

(c) How much time does it take for F to double?

2. Use a calculator where needed.

(a) $\ln(e^3) =$ (b) $e^{2.3} \approx$ (c) $\ln(5e^{2x}) =$

3. Each equation below is a best-fit line after log scaling. Turn it back into a model.

(a) If $\ln(y) = 0.04\,x + 2.3$, then $y \approx Ce^{kx}$ with $C =$ $k =$

(b) If $\ln(y) = 0.75\ln(x) + 4.2$, then $y \approx Cx^p$ with $C =$ $p =$